Data Visualization & Analytics Lab

Experiment No.: 1

To install any data visualization tool and create basic visualizations.

Experiment No. 1

1. **Aim:** To Install any data visualization tool (Tableau/PowerBI) and create basic visualizations (bar charts, line graphs, scatter plot, histogram and pie charts) using a sample dataset by applying effective visualization principles.

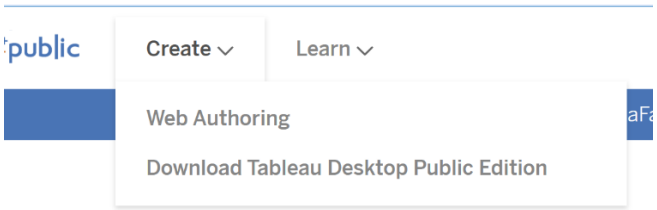
2. **Objectives:**
   - Develop skills to both design and critique visualizations.
   - To get familiar with data visualization tools.
3. **Course Outcomes:** Students will be able to understand the fundamentals of data visualization and use different data visualization tools
4. **Hardware / Software Required:** Tableau/PowerBI
5. **Theory:**
   - **Data visualization** is the graphical representation of information and data. By using visual elements like charts, graphs, and maps, data visualization tools provide an accessible way to see and understand trends, outliers, and patterns in data. Additionally, it provides an excellent way for employees or business owners to present data to non-technical audiences without confusion.
   - **Data visualization tools** provide the ability to see and understand data trends, outliers, and patterns in an easy, intuitive way. There are various data visualization tools available. One must choose the tool based on various factors such as its ease of use, types of graphical representations the tool can produce, size of the dataset the tool can handle etc. some of Data Visualization tools are Tableau, Power BI, Google Charts, JupyteR, Grafana etc.

Tableau

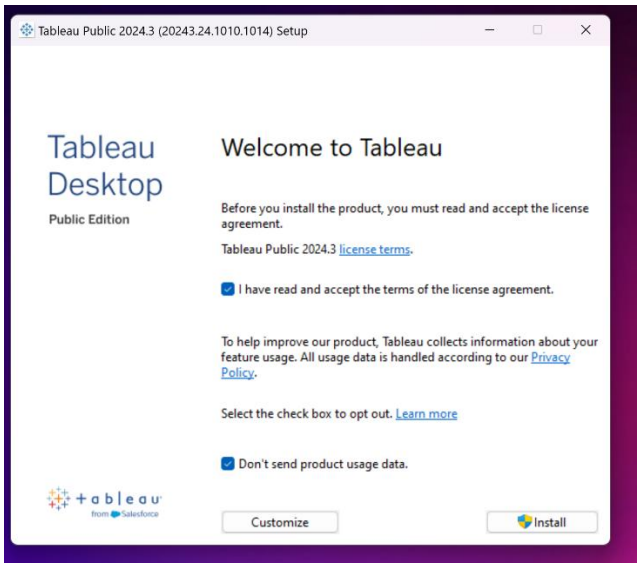
Tableau is a business intelligence and data visualization tool to get insights from data, find hidden trends, and make business decisions.

Install Tableau Public:

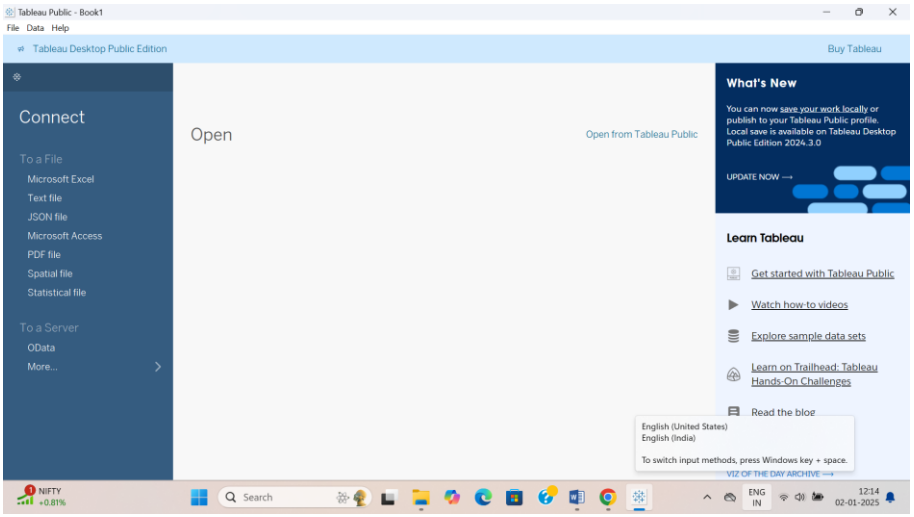
1. Go to the Tableau Public download page https://www.tableau.com/products/public
2. Enter your personal information
3. Confirm your email and sign in
4. Select Download Desktop Public edition from the Create drop-down menu



5. Click Download the App
6. Run the installer if it doesn't start automatically
7. Agree to the Public edition terms of service



The application will run automatically once installed.



Power BI:

Power BI is a data visualization tool that helps users analyze and visualize data from various sources. is a collection of software tools and connectors that help users turn data from different sources into

insights. It's designed for business intelligence purposes and can be used by professionals with varying levels of data knowledge.

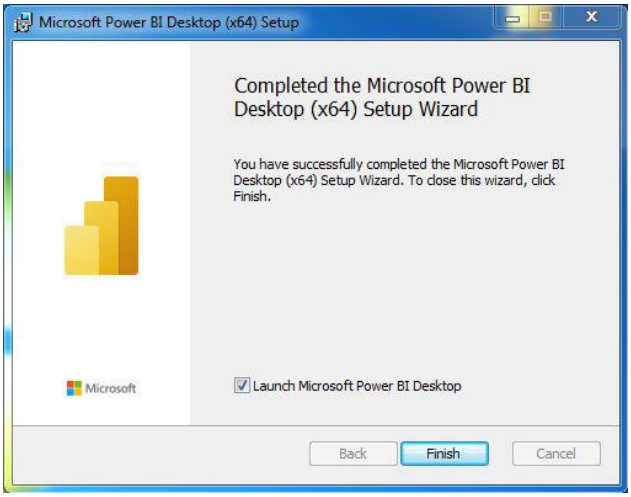
Power BI can connect to and analyze data from a variety of sources, including Excel spreadsheets, cloud-based data warehouses, and on-premises hybrid data warehouses. It can then visualize the data in a variety of styles, including graphs, maps, charts, and scatter plots.

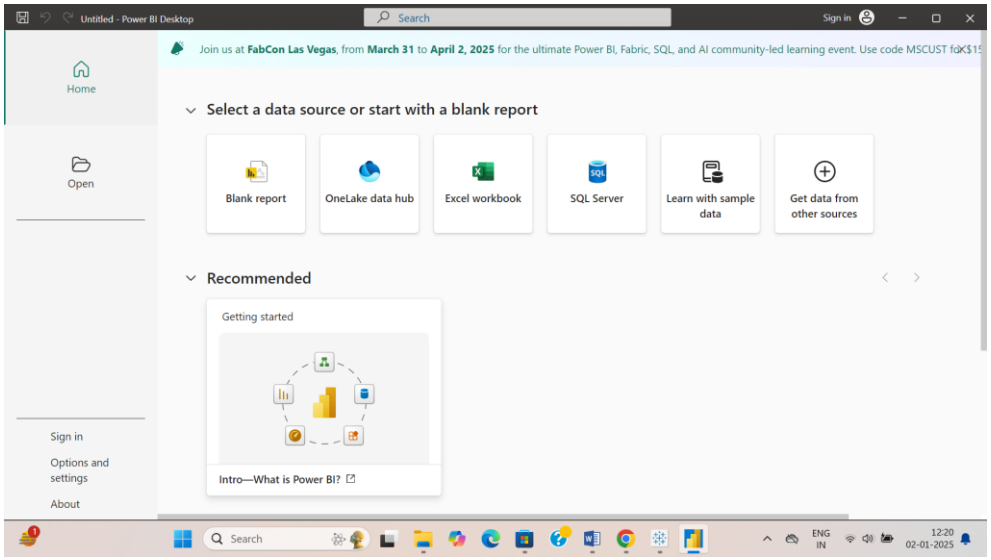
Power BI includes three components:

- Power BI Desktop: A free desktop app for transforming and visualizing data, and creating reports
- Power BI Service: An online service that allows users to update data in real time, publish and share it, and has similar capabilities to the desktop application
- Power BI Mobile: An app for viewing reports from iOS, Android, and Windows mobile devices

Install Power BI Desktop:

1. Go to the Power BI Desktop download page https://www.microsoft.com/en-us/power-platform/products/power-bi/desktop
2. Select “Download Free” and choose your preferred language.
3. Download the appropriate version for your system architecture (x86 or x64)
4. When the file has downloaded, open it, and use the setup wizard to install Power BI Desktop on your computer.

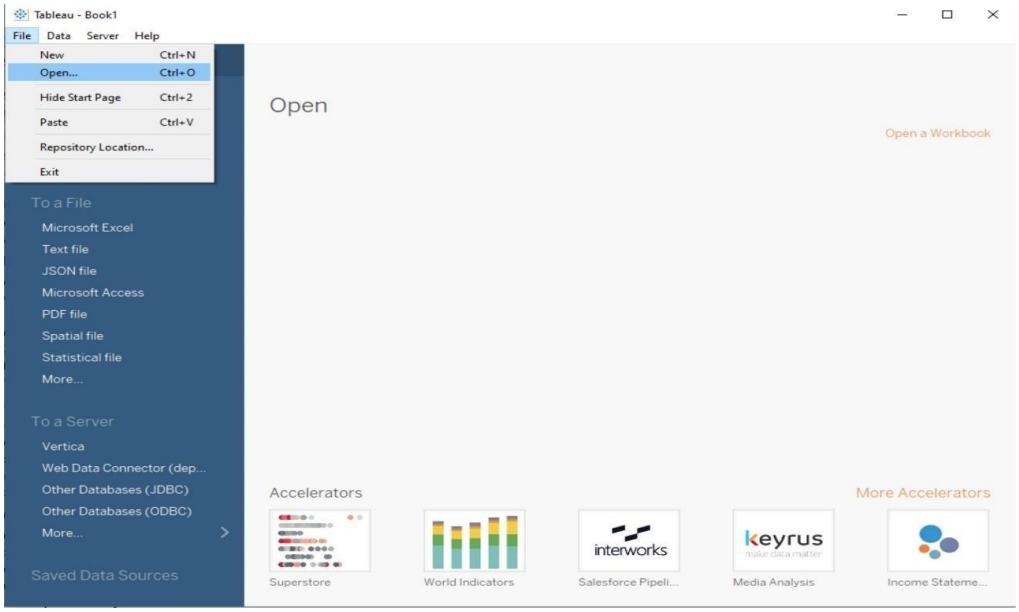




Creating Basic Visualization:

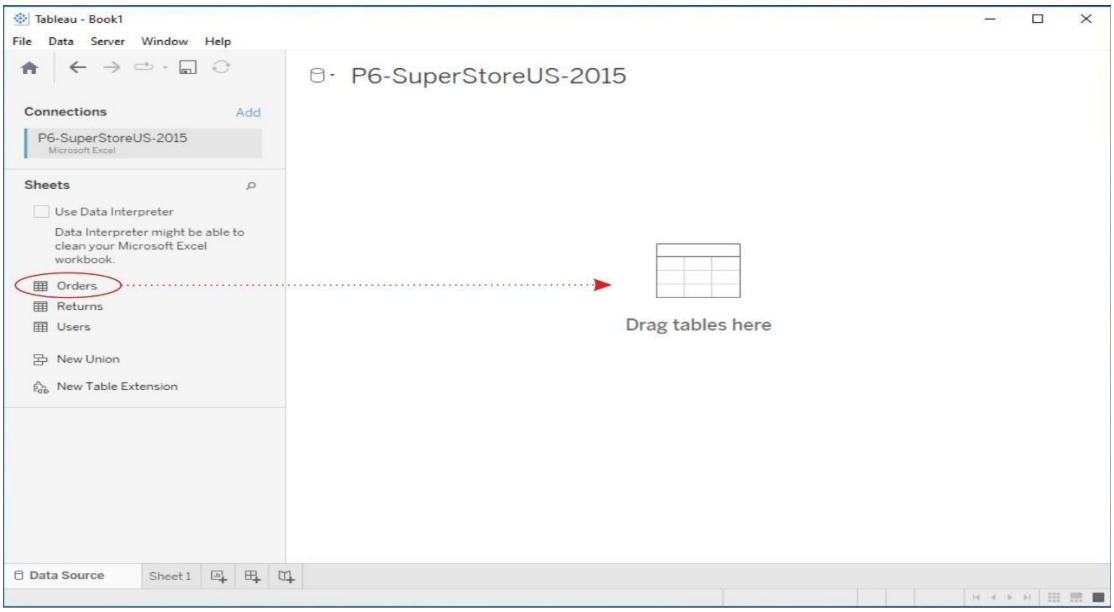
- **Implementation on Tableau:**

1. Connect Your Data to Tableau:
   - Launch Tableau Desktop.
   - Go to "File" Menu and then click on "Open".

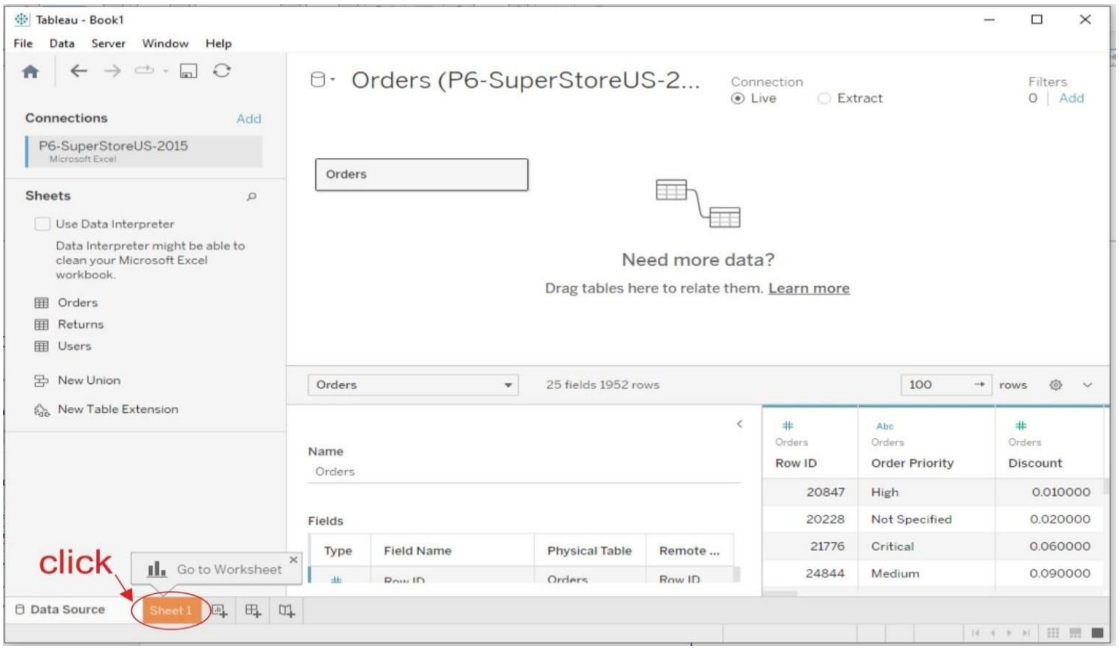


1.1 Choose the data source type (e.g., Excel, CSV, text file) and Select the data file(P6-SuperStoreUS-2015.xls) and click "Open".

1.2 Drag any table(e.g. Orders) into working area.

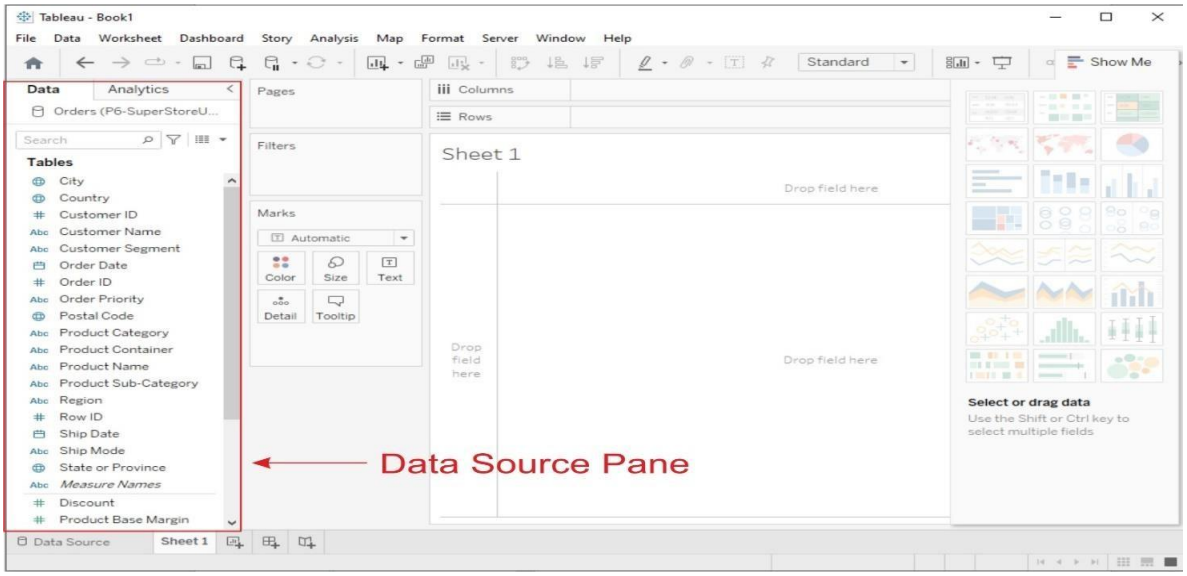


1.3 Click on Worksheet(**Sheet1**).



2. Data Source Pane:

Once your data is connected, the Data Source Pane will appear on the left-hand side of the Tableau interface. Here, you can see a preview of your data and perform data transformations or join multiple data sources if necessary.

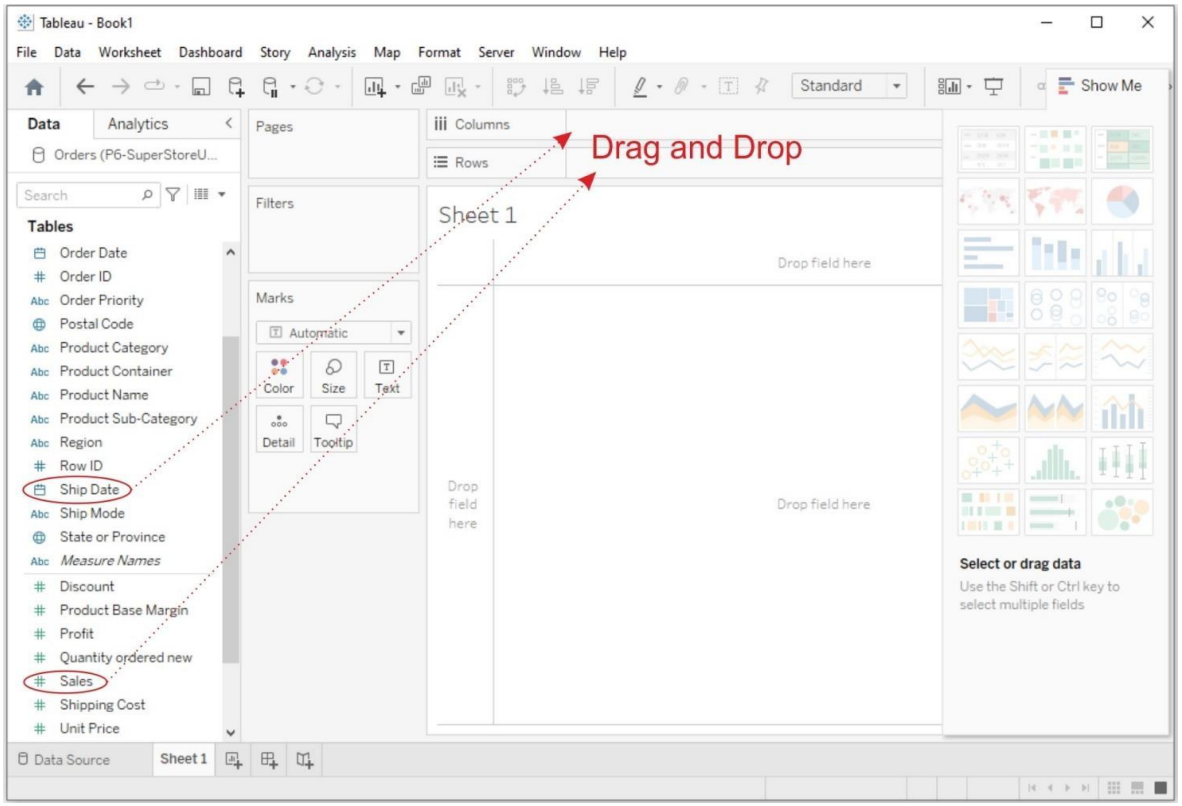


3. Creating Basic Charts:

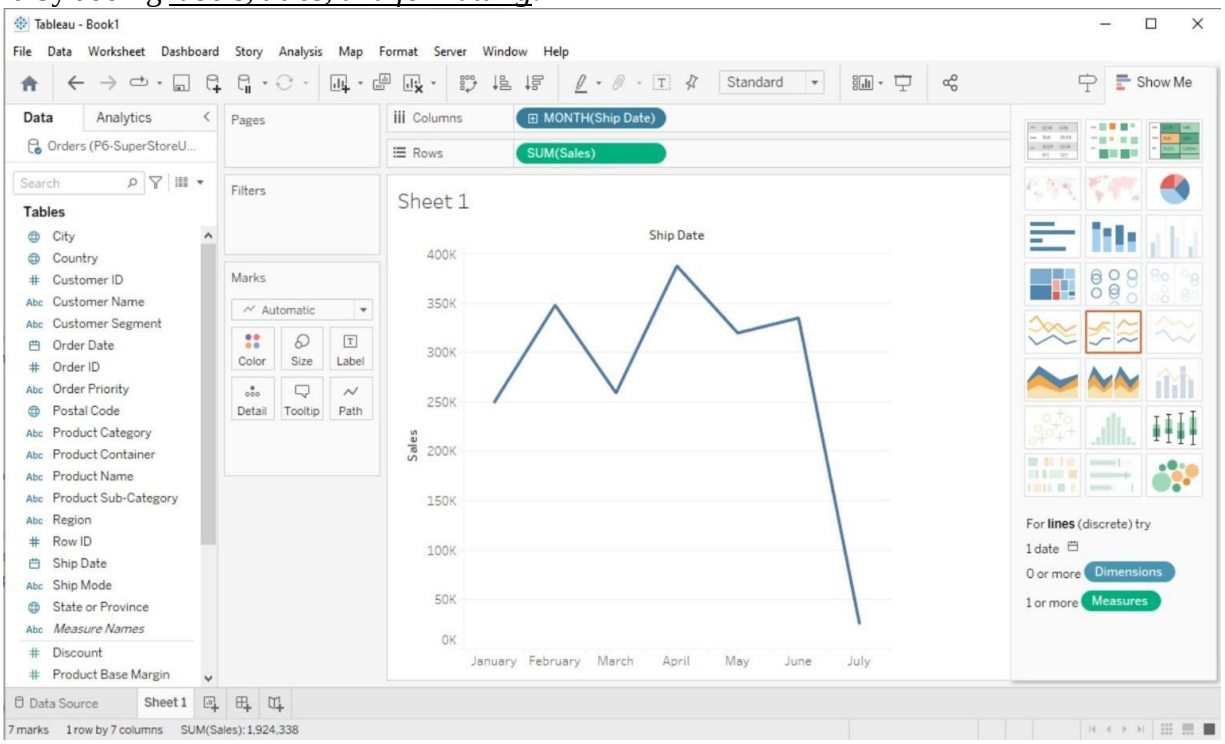
Now, let's create some basic charts using Tableau:

a. Line Chart:

1. From the "**Data Source pane**", drag and drop the date field to the **Columns shelf** and a numeric field (e.g., sales, revenue) to the **Rows shelf**.

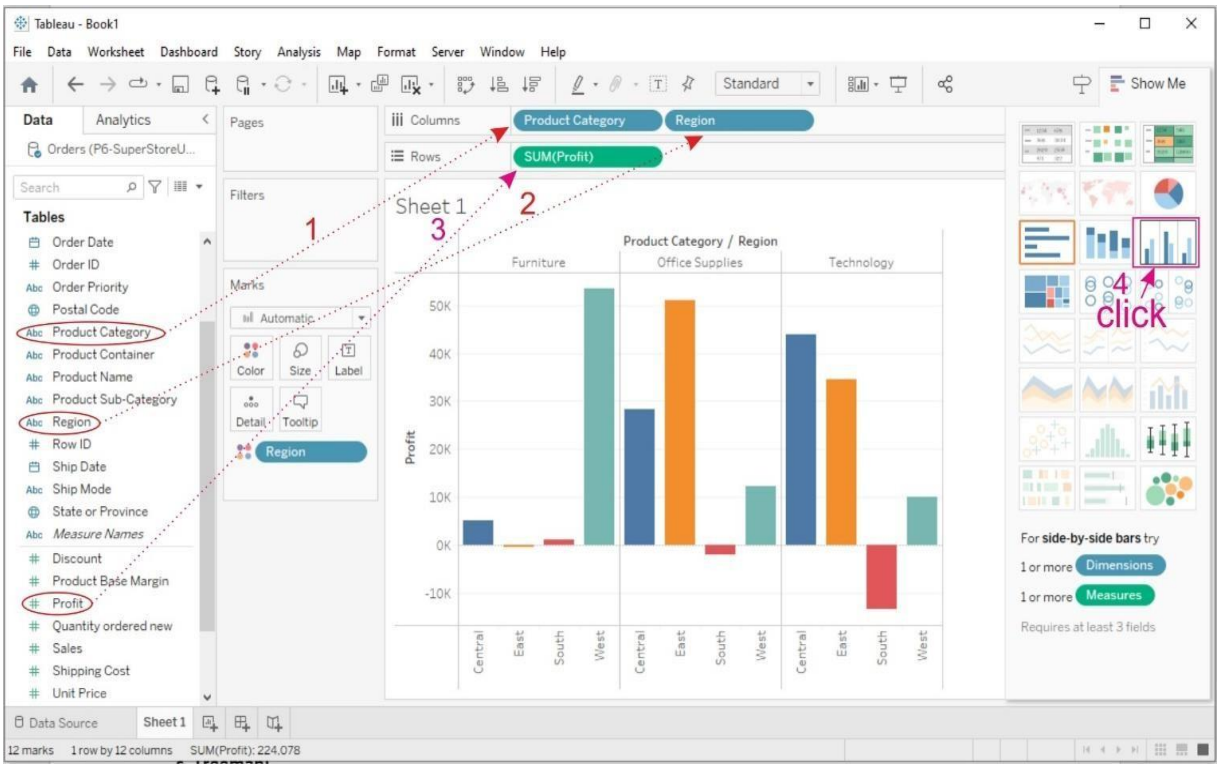


2. Then **Tableau** will automatically create a line chart. You can customize it by adding *labels, titles, and formatting*.

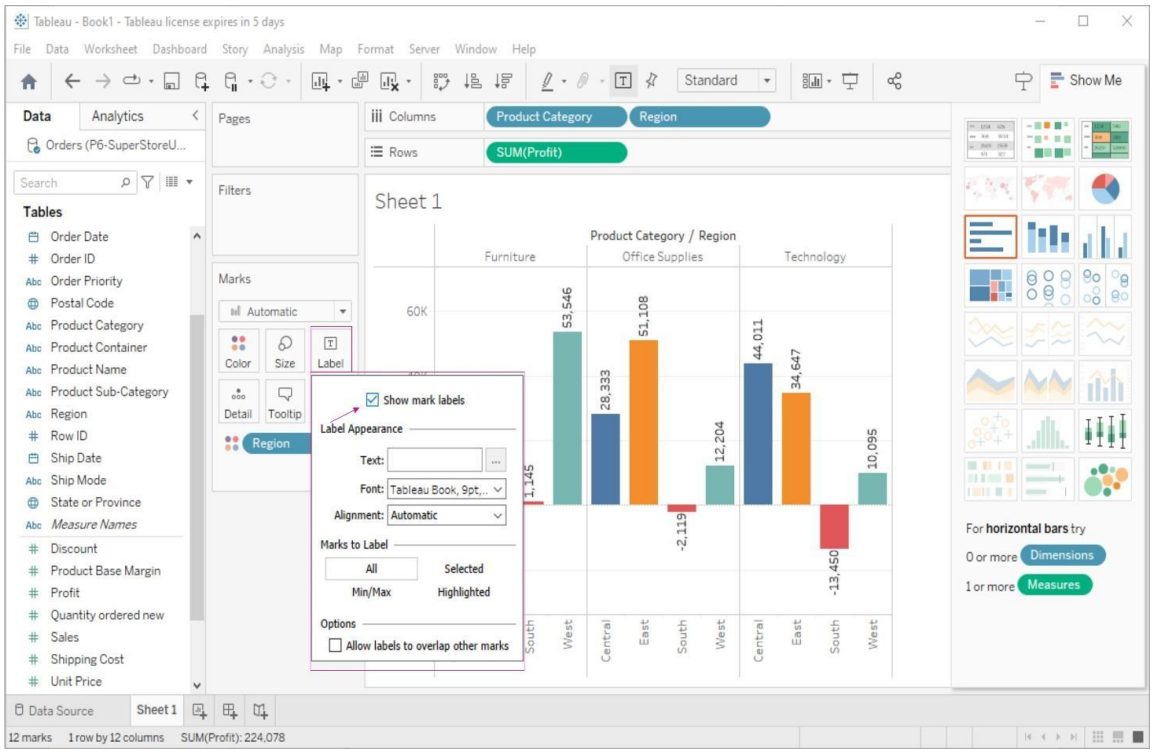


b. Bar Chart:

1. Drag and drop a **categorical field** (e.g., product category, region) to the Columns shelf and a **numeric field** to the Rows shelf.



2. Then **Tableau** will create a bar chart. You can adjust the orientation and formatting as needed. To display Labels on the bars click on Lables and select "**Show mark lables**"



6. Conclusion:

We have learnt about installation of the data visualization tools and studied their features. We created basic visualizations like bar chart, pie chart, line chart etc and understood the role of type of data in visualization

7. Viva Questions:

- What is data visualization? What are different visualization tools?
- What are different tools of data visualization?
- What are the basic visualization techniques?

References:

1. https://www.tableau.com/products/public
2. https://www.microsoft.com/en-us/power-platform/products/power-bi/desktop